ANDREW NEIL

AGENTIC AI ENGINEER | LLM APPLICATIONS | PYTHON AUTOMATION
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PROFILE

Computer Engineering student graduating in 2027 focused on building practical AI agents and automation systems. Hands-on experience with screenshot-driven planning and execution, tool routing, LLM provider integrations, workflow state, browser automation, and reliability safeguards. Strong Python and backend orientation.

AGENTIC AI & TECHNICAL SKILLS

Agentic AI: LLM planning, tool calling, multi-agent routing, structured JSON actions, workflow state, action verification, bounded retries
LLM & automation: Gemini, provider-configurable LLM gateways, local-model integration, LangChain, Playwright, Selenium, PyAutoGUI
Python & backend: Python, Django, FastAPI, REST APIs, pytest, SMTP, Git/GitHub, Linux/CLI
Computer vision & data: YOLOv8, OpenCV, OCR/ANPR, Pandas, NumPy, scikit-learn, PyTorch, pandas-ta, Parquet

EXPERIENCE

Backend Engineer Intern | 41studio • *Aug 2026 – Present*
Contributing to a Django ERP for a polytechnic college, supporting student records and academic workflows including marks entry, attendance, marksheets, and certificate generation across institutional roles.

Freelance Python Developer | Upwork · 3 completed contracts • *2023–2025*
Built a website appointment monitor with email alerts through a client-provided SMTP server; developed a LangChain CLI that generates three LLM response variants with regeneration and selected-output context; delivered Selenium-based browser automation and scraping. Earned two 5-star client reviews.

SELECTED PROJECTS

OSCAR — Multi-Agent AI Desktop Automation Assistant
Tech: *Python, Flutter/Dart, Gemini, Playwright, PyAutoGUI, YOLOv8/OCR, pytest*
- Built a Windows-first assistant that turns natural-language requests into workflows across browser, desktop UI, and terminal agents, coordinated by a Python orchestration backend and Flutter desktop UI.
- Implemented screenshot-driven planning: the model proposes structured JSON actions, Python dispatches mouse/keyboard/scroll inputs, and fresh observations inform the next step. Added record/replay workflows and persistent execution state.
- Improved execution safeguards with tool-payload validation, outcome checks, bounded retries, duplicate-typing prevention, and explicit completion checks; added a fast path for simple tasks to avoid unnecessary LLM calls.
- 336 automated regression tests pass; the suite uses mocked Windows APIs and does not establish real-world click accuracy. Repository: github.com/smmsa1912/OSCAR

TradeAssist — Quantitative Trading Research & Risk Management
Tech: *Python, Angel One SmartAPI, Pandas, NumPy, pandas-ta, Parquet*
- Built historical market-data ingestion and backtesting with technical signals, trade simulation, transaction costs, 5-bps slippage, stop/trailing-stop logic, portfolio metrics, buy-and-hold comparisons, and 13-fold walk-forward evaluation.
- Best recorded IRFC SMA 20/50 crossover + 5% stop-loss result (Jan 2023–Sep 2026): +24.17% total return, 6.75% CAGR, and −11.03% max drawdown over 86 trades. Historical backtest, not live performance.

FinVerse — AI-Powered Blockchain Financial Platform
Tech: *React, Django, Flutter, Python, blockchain, ESP32, NFC/RFID*
- Built a working prototype spanning a React dashboard, Django backend, Flutter app, AI-assisted financial workflows, blockchain functionality, and ESP32/NFC-based hardware authentication.

Selected Additional Projects
Tech: *Computer vision, digital identity, RFID*
- Dynamic Toll Charging System — combined YOLOv8 vehicle detection, OCR/ANPR, RFID-based FASTag simulation, and a dummy wallet; won the college mini-project competition.
- KYC Smart Card — co-developed a DigiLocker-inspired document-storage and sharing prototype with Django backend and RFID hardware; documented in IJIRT papers.

EDUCATION & PUBLICATIONS

B.E. Computer Engineering, Xavier Institute of Engineering (XIE), Mumbai | Expected 2027 | CGPA: 7.48/10 through Semester VI
Publications: Dynamic Toll Charging System, IJIRT (2026) — ijirt.org/article?manuscript=191077; KYC Smart Card research papers, IJIRT (2024/2025).